# Trading Scope for Credibility in Difference-in-Differences

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**Abstract**

Difference-in-differences practitioners routinely confront parallel trends failing for some treated cohorts but not others. Existing responses keep the average treatment effect on the treated (ATT) as the target. Heterogeneity-robust estimators point-identify it while maintaining parallel trends, and sensitivity methods bound it, so both are compromised, through bias or a wide set, exactly when some cohorts offend. We propose a different response: change the estimand. We define a credible-subpopulation local ATT (LATT), the treatment effect for the subpopulation of cohorts whose parallel trends assumption is credible, and show it is point-identified precisely when the full ATT is not. Estimation is a reweighting of standard group-time effects toward credible cohorts, combined with honest sensitivity bounds on the residual violation of the selected set. The method's advantage over the ATT grows with how informative pre-trends are about post-treatment violations. Simulations confirm that the LATT recovers a clean effect where the ATT estimator is biased and that honest sensitivity bounds restore coverage where point estimates collapse. Two staggered applications, Medicaid expansion and Walmart entry, show the method recovering a point-identified effect that survives honest sensitivity analysis and de-attenuating a pooled estimate biased by endogenous adoption.

**Keywords:** Difference-in-differences, Staggered adoption, Parallel trends, Robust inference, Sensitivity analysis, Partial identification.

**JEL classification:** C1, C23, C51.

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# 1 Introduction

Difference-in-differences is among the most widely used research designs in empirical economics. Its credibility rests on the parallel trends assumption: absent treatment, the average outcomes of treated and comparison units would have evolved in parallel. Researchers are rarely certain that this holds, and it has become standard practice to assess it by testing for pre-treatment differences in trends.

That practice is fragile. Pre-trends tests are often underpowered against economically meaningful violations (Bilinski and Hatfield, 2020; Freyaldenhoven et al., 2019; Kahn-Lang and Lang, 2020; Roth, 2022). Conditioning the analysis on passing them induces a pre-test selection bias (Roth, 2022). And even a clean pre-period does not guarantee a clean post-period, since nothing prevents a confound from arriving at the moment of treatment (Kahn-Lang and Lang, 2020).

A large recent literature, surveyed in Roth et al. (2023) and de Chaisemartin and D'Haultfœuille (2023), has responded by relaxing parallel trends itself. Building on Manski and Pepper (2018), Rambachan and Roth (2023) abandon exact parallel trends and instead bound how different the post-treatment violations can be from the observed pre-trends, reporting a uniformly valid confidence set for the ATT in place of a point estimate. A Bayesian variant (Kwon and Roth, 2024) instead places a prior on the violation $\delta$, calibrated from the pre-trends by empirical Bayes, to sharpen the set into a posterior mean and credible interval. Both keep the ATT as the target and use the pre-trends to extrapolate the violation across periods. Our method uses them to select across cohorts. In non-staggered designs with a long pre-period, de Chaisemartin (2026) instead ranks the post-treatment difference-in-differences against the distribution of the pre-treatment ones, replacing parallel trends with a stability restriction.

The ATT is where a structural limitation appears. When parallel trends fails for some cohorts but holds for others, the ATT, an average over all treated cohorts including the offending ones, is precisely the object that is hard to recover. A point estimate of it is then biased, and the sensitivity set, though honest, can be wide and uninformative, because it must accommodate the worst offenders in the average.

This paper takes a different route, familiar from other corners of causal inference: when the population target is not credibly identified, retreat to a subpopulation where it is. This is the logic of the local average treatment effect (Imbens and Angrist, 1994), of the optimal-subpopulation average treatment effect under limited overlap (Crump et al., 2009), and of overlap weighting (Li et al., 2018). In each case one trades the original estimand for a credibly-identified effect on a selected subpopulation, and defends the trade by characterizing whom the subpopulation comprises.

We develop the staggered-DiD instance of this principle. We define the credible-subpopulation LATT, the weighted average of group-time effects restricted to cohorts whose parallel trends assumption is credible, and show that it is point-identified even when the full ATT is not, because the violations of the excluded cohorts do not enter it. Estimation is a transparent reweighting of the group-time effects of Callaway and Sant'Anna (2021), which we adopt for concreteness. Because the method only restricts the aggregation to a subset of cohorts, it applies to any heterogeneity-robust estimator that delivers group-time effects.[1] The credibility of a cohort is assessed from its pre-trends, so that the reweighting is a data-dependent selection, and we treat inference on the selected set with the sensitivity machinery of Rambachan and Roth (2023).

The method thus occupies a distinct point on a frontier trading off credibility, precision, and scope. Relative to a heterogeneity-robust point estimate of the ATT, it targets a subpopulation to buy back credibility. Relative to a HonestDiD set for the ATT, it recovers a sharper, point-identified answer to a narrower question.

Its advantage is governed by how informative pre-trends are about post-treatment violations. When they are uninformative, selecting on them buys nothing and the identifying assumption must rest on economic context. In that regime the natural relative-magnitudes sensitivity restriction is self-undermining. Anchoring its bounds to the observed pre-trends, it reports false precision when those are flat, whereas a smoothness restriction, keyed to structure rather than magnitude, is not.

The remainder of the paper is organized as follows. Section 2 fixes the staggered-DiD setup, defines the credible-subpopulation LATT, and develops its identification, estimation, and sensitivity-based inference. Section 3 reports the simulation study, Section 4 applies the method to Medicaid expansion and Walmart entry, and Section 5 concludes.

# 2 The credible-subpopulation LATT

## 2.1 Setup

We adopt the staggered adoption framework of Callaway and Sant'Anna (2021). There are periods $t = 1, \ldots, T$ and units $i$ indexed by their first treatment date $G_i \in \{2, \ldots, T\} \cup \{\infty\}$, where $G_i = \infty$ denotes never-treated. Let $Y_{it}(g)$ denote the potential outcome in period $t$ if unit $i$ is first treated in period $g$, and $Y_{it}(\infty)$ the never-treated potential outcome. The building-block causal parameter is the group-time average treatment effect on the treated,

$$\mathrm{ATT}(g,t) = \mathbb{E}[\, Y_{it}(g) - Y_{it}(\infty) \mid G_i = g \,]\,. \tag{1}$$

Summary parameters are weighted aggregations $\theta = \sum_g \sum_t w(g,t)\,\mathrm{ATT}(g,t)$ with researcher-chosen weights (Callaway and Sant'Anna, 2021), of which the overall ATT and the event-study path are special cases.

It is convenient to work with the event-study representation and the decomposition of Rambachan and Roth (2023). Let $\hat{\beta} = (\hat{\beta}'_{\text{pre}}, \hat{\beta}'_{\text{post}})'$ be an asymptotically normal vector of event-study coefficients estimated by any of the heterogeneity-robust procedures above, with $\sqrt{n}(\hat{\beta} - \beta) \to \mathcal{N}(0, \Sigma)$. The

[1] These estimators (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024; de Chaisemartin and D'Haultfœuille, 2020) were developed to repair the aggregation failure of two-way fixed effects under staggered adoption, whose implicit weights on group-time effects can turn negative when treatment effects are heterogeneous (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020; Sun and Abraham, 2021). That repair concerns aggregation rather than identification: these estimators maintain parallel trends and are biased when it fails, which is the failure we address.

estimand decomposes as

$$\beta = \tau + \delta, \qquad \tau_{\text{pre}} = 0, \tag{2}$$

where $\tau$ collects the causal effects (zero before treatment, by no anticipation) and $\delta$ is the differential trend between treated and comparison groups that would have occurred absent treatment. Parallel trends is the restriction $\delta_{\text{post}} = 0$, under which $\beta_{\text{post}} = \tau_{\text{post}}$, and a pre-trends test is a test of $\delta_{\text{pre}} = 0$.

We will require the analogous objects at the cohort level. Write $\delta_g$ for the differential trend of cohort $g$ and say that parallel trends holds for $g$ if $\delta_{g,\text{post}} = 0$. A key feature of applications is that this may hold for some cohorts and fail for others. Defending the timing of treatment as unrelated to the outcome path is already delicate in a two-group design, and staggered adoption raises the burden rather than lowering it, since each cohort's timing must be defended on its own: one cohort's adoption may be as good as randomly timed while another's coincides with a confounding shock.

Finally, we distinguish two regimes for how cohorts are judged credible, because they carry different inferential guarantees. Under ex-ante (or independent) selection, credibility is decided from pre-determined covariates, institutional knowledge, or an independent data split, so that the selected set is statistically independent of the estimation errors in $\hat{\beta}$. Under data-driven selection, credibility is assessed from the realized pre-trends $\hat{\beta}_{\text{pre}}$, so that the selected set depends on the estimation sample. The distinction is immaterial for the estimand and its identification, but central for inference, and we treat it explicitly below.

## 2.2 Estimand and identification

Let $S$ denote a set of cohorts judged to have credible parallel trends, produced by a selection rule $R$ that we specify below. We define the target as the treatment effect for that subpopulation.

**Definition 1** (Credible-subpopulation LATT)**.** *For cohort weights $w_g > 0$ (e.g. proportional to cohort size), the credible-subpopulation local ATT is*

$$\theta_S \;=\; \frac{\sum_{g\in S} w_g \,\mathrm{ATT}(g,\cdot)}{\sum_{g\in S} w_g}, \tag{3}$$

*where $\mathrm{ATT}(g,\cdot)$ is any within-cohort aggregation of interest (a fixed event-time, or an average over post-treatment periods).*

The value of the target is that it is identified under a weaker condition than the ATT.

**Proposition 1** (Identification)**.** *If parallel trends holds for every $g \in S$, i.e. $\delta_{g,\text{post}} = 0$ for all $g \in S$, then $\theta_S$ is point-identified by the reweighted group-time effects, $\theta_S = \big(\sum_{g\in S} w_g\, \beta_{g,\text{post}}\big)/\sum_{g\in S} w_g$, and this holds irrespective of whether $\delta_{g,\text{post}} \neq 0$ for cohorts $g \notin S$.*

*Proof.* For $g \in S$, $\beta_{g,\text{post}} = \tau_{g,\text{post}} + \delta_{g,\text{post}} = \tau_{g,\text{post}}$ by the hypothesis and (2). The reweighted sum over $S$ therefore equals the corresponding weighted average of causal effects, which is $\theta_S$ by definition. Cohorts outside $S$ receive zero weight, so their differential trends do not enter. $\square$

Proposition 1 is elementary, but it is the paper's central point. By changing the target to the credible subpopulation, one converts a partially-identified problem (the ATT, when some cohorts violate parallel trends) into a point-identified one, at the cost of answering a narrower question. This is exactly the trade made by Imbens and Angrist (1994), Crump et al. (2009), and Li et al. (2018) in other settings, and it carries the same interpretive obligation: one must characterize the subpopulation $S$ that the estimand describes.

The construction uses staggered timing only to supply the groups. What Proposition 1 requires is a partition of the treated population into groups whose parallel trends can each be assessed, together with the event-study structure that makes such assessment possible. Adoption cohorts are the natural groups under staggered timing, and their credibility heterogeneity comes for free from the variation in when and why units adopt. In a non-staggered design, where all treated units share one treatment date, the same construction applies with groups defined by covariates or geography, provided two conditions that staggering would otherwise supply are argued rather than assumed: the groups must differ in the credibility of their parallel trends, and there must be enough pre-periods to estimate each group's pre-trend rather than chase noise. The groups should accordingly be subpopulations large enough to assess, not individual units, whose pre-trends are too noisy to select on without reintroducing the pre-test bias in its most severe form. Read this way, the method is the parallel-trends analogue of selecting on overlap (Crump et al., 2009; Li et al., 2018): one trims to the subpopulation on which the identifying assumption is credible, using pre-trends in place of the propensity score.

### 2.3 Selecting the credible cohorts

The selection rule $R$ maps the data to the set $S$, and its form fixes both which subpopulation $\theta_S$ describes and the inferential guarantees available. It takes two forms, matching the two regimes of Section 2.1. Under ex-ante selection, $S$ is fixed before estimation from pre-determined covariates, institutional knowledge of which cohorts were plausibly confounded, or an independent data split. Under data-driven selection, credibility is read from the estimated pre-trends: a cohort enters $S$ when its estimated pre-trend has small curvature,

$$S \;=\; \Big\{ g : \max_{e<0} \big|\hat{\beta}_{g,\text{pre}}(e{+}1) - 2\,\hat{\beta}_{g,\text{pre}}(e) + \hat{\beta}_{g,\text{pre}}(e{-}1)\big| \le c \Big\}, \tag{4}$$

for a threshold $c$, with the reference period $\hat{\beta}_g(0) = 0$ supplying the boundary term.

The choice of statistic is not incidental. It is dictated by the honest sensitivity analysis of the next subsection, which bounds the residual violation by its curvature, the second difference of the differential trend, so the screen is posed in the same currency: it drops cohorts whose pre-trend bends rather than those whose pre-trend is merely large or steeply sloped. Under $\Delta^{SD}(M)$ a level shift or a linear drift is harmless, since the class constrains only the second difference and the associated interval extrapolates a linear pre-trend before differencing it out, so neither survives into the honest bound on $\theta_S$. Curvature is the one feature the interval cannot absorb, which is why the screen is keyed to it, and keying selection to the same functional the restriction class bounds makes the two stages consistent: a stricter $c$ then removes exactly the residual curvature that a smaller $M$ must later dominate, and a class built to treat a different feature as the violation would call for a correspondingly different screen.

The threshold trades scope against credibility: a lax $c$ retains more cohorts but admits larger residual violations, a strict $c$ the reverse. Because a cohort's pre- and post-treatment coefficient errors are correlated in finite samples, rule (4) makes $S$ random and dependent on the estimation sample, which is the source of the pre-test bias below and of the inference subtleties that follow it.

We do not recommend committing to a single $c$. Because the sensitivity interval of Section 2 is applied to whichever set $c$ selects and must dominate that set's residual pre-trend, a stricter $c$ yields a cleaner set that a smaller bound $M$ suffices to cover, while a laxer $c$ buys scope at the cost of a larger $M$. The threshold therefore trades scope for precision rather than credibility for nothing, and the honest object to report is the estimand together with its interval traced as a function of $c$, the selection analogue of the breakdown curve we report for $M$. A reader then sees the scope-credibility-precision frontier directly and need not trust a single cutoff. When a single value is nonetheless wanted, we recommend calibrating it to negligibility rather than to significance: set $c$ to the largest pre-trend that would bias $\theta_S$ by less than a stated tolerance, in the spirit of the non-inferiority tests of Bilinski and Hatfield (2020), rather than to a critical value of a pre-trends test, which would reinherit the low power that motivates the paper. The Walmart application of Section 4 reports such a path.

### 2.4 Estimation

The estimator $\hat{\theta}_S$ replaces population objects by their Callaway and Sant'Anna (2021) sample analogues and reweights over the selected cohorts.

**Proposition 2** (Consistency and efficiency)**.** *Under ex-ante selection and the regularity conditions of Callaway and Sant'Anna (2021), $\hat{\theta}_S$ is consistent for $\theta_S$. If parallel trends holds for all cohorts, $\hat{\theta}_S$ is consistent for the ATT, with an efficiency loss relative to estimators that use all cohorts.*

Under data-driven selection the rule (4) couples $S$ to the estimation errors: a confounded cohort whose pre-trend noise happens to fall below $c$ is admitted, and its differential trend enters $\hat{\theta}_S$. This is a pre-test selection bias in the sense of Roth (2022), with a definite source and a definite fate.

**Proposition 3** (Pre-test bias)**.** *Under data-driven selection, $\hat{\theta}_S$ carries a finite-sample bias equal to the weighted average of the admitted confounded cohorts' post-treatment differential trends, times their probability of admission. As per-cohort information grows this admission probability tends to zero, so the bias vanishes and $\hat{\theta}_S$ is consistent for $\theta_S$, though not finite-sample unbiased.*

### 2.5 Honest inference on the selected set

Selecting credible cohorts does not by itself deliver parallel trends: a cohort with flat pre-trends may still have a nonzero post-treatment differential trend, and selection removes gross offenders but not subtle residual violations. We therefore accompany the point estimate with the sensitivity analysis of Rambachan and Roth (2023), applied to the selected-cohort aggregate event study: one imposes a restriction $\delta_{S,\text{post}} \in \Delta$ on the residual differential trend and reports the implied confidence set for $\theta_S$. We adopt the smoothness class $\Delta^{SD}(M) = \{\delta : |\delta_{t+1} - 2\delta_t + \delta_{t-1}| \leq M \ \forall t\}$, which admits the near-optimal fixed-length confidence intervals (FLCIs) of Armstrong and Kolesár (2018).

Validity depends on the selection regime. Under ex-ante selection $S$ is independent of $\hat{\beta}$, so the uniform validity of Rambachan and Roth (2023) applies verbatim. Under data-driven selection $S$ is random, and two issues arise. The first is post-selection inference: when cohorts are well separated from the threshold, selection is asymptotically harmless.

**Proposition 4** (Selection consistency and oracle coverage)**.** *Suppose the statistic on which selection is based is consistent for its population counterpart, and that the following separation condition holds: there exists $\epsilon > 0$ such that, for every cohort, the population value of that statistic lies at distance at least $\epsilon$ from the threshold $c$. Then $P(\hat{S} = S^*) \to 1$, where $S^*$ denotes the population credible set, and the FLCI computed on the estimated set $\hat{S}$ has the same asymptotic coverage as the oracle interval computed on $S^*$, and in particular attains its nominal level.*

*Proof.* By consistency of the selection statistic together with the separation condition, the probability that any given cohort is misclassified tends to zero. Since the number of cohorts is finite, a union bound gives $P(\hat{S} \neq S^*) \to 0$. On the complementary event the estimator, its covariance estimate and the resulting interval coincide exactly with their oracle counterparts computed on the non-random set $S^*$. The two sequences therefore share a limiting distribution, and the coverage of the interval computed on $\hat{S}$ converges to that of the oracle interval, which is nominal by the validity of Rambachan and Roth (2023) applied to a fixed set. $\square$

Two caveats attach to Proposition 4. First, the separation condition is untestable, and is nearly the parallel trends question in disguise: to assert that every cohort is plainly credible or plainly not is close to asserting one already knows which cohorts are credible, so the result relocates the identifying content rather than supplying it. Second, the guarantee is pointwise, not uniform: Leeb and Pötscher (2005) show that a post-selection estimator's distribution cannot be estimated uniformly consistently, so naive post-selection intervals lack uniform validity (Leeb and Pötscher, 2008). The obstruction is confined to local-to-threshold configurations, where the selection statistic drifts toward the boundary at the sampling rate. Away from the boundary, selection is asymptotically deterministic and inference behaves conventionally. A researcher unwilling to assume separation can sidestep both caveats by selecting on information independent of the estimation errors, such as pre-determined covariates, a sample split, or randomized (data-carving) selection (Fithian et al., 2014). These buy validity through independence between selection and estimation rather than correct selection, at some cost in efficiency.

The second issue is that the bound $M$ must accommodate a random selected set. The FLCI at $M$ is valid only if the selected set's residual curvature lies in $\Delta^{SD}(M)$. Since that curvature is itself random, unconditional coverage requires $M$ to dominate it with high probability.

**Lemma 1** (Calibration to the random selected set)**.** *Under data-driven selection, choosing $M$ equal to the mean residual curvature of the selected aggregate undercovers. Unconditional coverage requires $M$ to exceed the mean and to approach an upper quantile of the residual-curvature distribution, with the FLCI's sampling slack providing a partial buffer.*

The practical implication is that $M$ should be calibrated to the residual violation of a typical selected set, not to its mean.

Finally, we caution against one otherwise-natural choice of $\Delta$. The relative-magnitudes restriction $\Delta^{RM}(\bar{M})$ of Rambachan and Roth (2023), following Manski and Pepper (2018), bounds post-treatment violations by $\bar{M}$ times the observed maximal pre-treatment violation. Because it anchors its bound to the observed pre-trends, it is self-undermining precisely in the regime where our method's value is in question.

**Proposition 5** (Relative magnitudes is self-undermining)**.** *The identified-set half-width under $\Delta^{RM}(\bar{M})$ is proportional to the observed pre-trend magnitude. When pre-trends are uninformative about post-treatment violations, the observed pre-trends are small while the true violation need not be, so $\Delta^{RM}(\bar{M})$ produces intervals whose width tends to zero even as the bias does not, yielding undercoverage. A structural restriction such as $\Delta^{SD}(M)$ does not share this defect, since its bound keys off smoothness rather than pre-trend magnitude.*

# 3 Simulation study

## 3.1 Design

We study the method in two environments. In the reduced-form model we draw event-study coefficients directly from their asymptotic distribution, $\hat{\beta} \sim \mathcal{N}(\tau + \delta, \Sigma)$ with $\Sigma$ known, which isolates the identification and inference mechanics in the setting where the underlying theory is exact. In the panel model we generate individual outcomes for a staggered design and estimate both the coefficients and their covariance from the data, which confirms that the conclusions survive realistic estimation. Unless stated otherwise, results are averages over eight to twenty thousand replications.

The cohort structure is the same throughout. There are twelve cohorts, of which some are clean, with $\delta_g = 0$, and the remainder confounded, with a nonzero differential trend. Every cohort is given the same true treatment effect, normalized to one. This normalization is deliberate: it makes the true ATT and the true LATT both equal to one, so that any departure of an estimator from one is attributable to a violation of parallel trends rather than to treatment-effect heterogeneity. It therefore isolates the identification problem that the paper is about.

Throughout, each cohort carries a four-period pre-treatment and four-period post-treatment event study with the reference period normalized to zero. A confounded cohort carries a curved differential trend: its post-treatment curvature is $C > 0$, the maximal second difference of $\delta_{g,\text{post}}$, and its pre-treatment curvature is $\phi C$, for a scalar $\phi \geq 0$ we call the informativeness parameter and that is the master axis of the study. Concretely $\delta_e = (\phi C/2)e^2$ before treatment and $(C/2)e^2$ after, so the post-period bend is foreshadowed in the pre-period to the degree $\phi$. When $\phi = 0$ the confound is flat before treatment and no pre-trend rule can detect it; as $\phi$ grows it announces itself and becomes detectable. Because $C$ is the post-treatment curvature it is directly comparable to the smoothness bound $M$ of $\Delta^{SD}(M)$, which makes the validity statement "covers if and only if $M \geq C$" checkable, and it makes $\phi$ the operational version of the informativeness of pre-trends discussed in Section 2.

The selection rule is data-driven and posed in the same curvature currency as the honest inference: cohort $g$ is retained if its estimated pre-trend has small curvature, $\max_e|\hat{\beta}_{g,\text{pre}}(e+1) - 2\hat{\beta}_{g,\text{pre}}(e) + \hat{\beta}_{g,\text{pre}}(e-1)| \leq c$. Because a second difference is a noisier functional than a level, this screen needs more per-cohort precision than a magnitude screen to select as reliably, a cost we make explicit below. The estimator of the ATT averages the post-treatment effect over all cohorts, as a heterogeneity-robust estimator would, and the LATT averages it over the selected cohorts only. Because the sampling errors of a cohort's pre- and post-treatment coefficients are correlated, the selection is genuinely data-dependent in the sense of Section 2, and the pre-test bias of Proposition 3 is operative.

### 3.2 The estimand: recovering a clean effect where the ATT is biased

We first illustrate Proposition 1 in the simplest configuration, with a subset of cohorts violating parallel trends in the post-period. Table 1 reports the result. The estimator of the ATT is biased by +0.438: it averages over all cohorts, so the offending cohorts' differential trends enter it directly. The LATT lands at 1.000, on the truth, because the selection removes those cohorts from the average and, by Proposition 1, their violations then do not enter the estimand at all.

The lower panel of the table sweeps the sampling noise $s$, and is the more informative comparison. As precision improves the LATT's bias falls monotonically from +0.137 to zero: this residual is the pre-test bias of Proposition 3, which arises because the selection is made on the same noisy pre-trends used for estimation, and it vanishes as the selection rule becomes reliable. The ATT estimator's bias, by contrast, is frozen at +0.438 regardless of precision. The contrast is the paper's basic point in miniature: the LATT faces a noise problem, which more data cures, whereas the ATT estimator faces a wrong-target problem, which no amount of data cures.

### 3.3 The scope condition

Figure 1 traces the method across the informativeness axis $\phi$ described in Section 3.1, and its four panels should be read together.

The upper-left panel plots the bias of the two point estimators against the truth. The bias of the ATT estimator is a flat line at +0.66: it never consults pre-trends, so its performance cannot depend on how informative they are. The LATT's bias begins on top of that line at $\phi = 0$ and falls to zero as pre-trends become informative. The vertical gap between the two curves is therefore exactly the method's advantage, and reading it off from left to right gives the scope condition: the advantage is

Table 1: The estimand demonstration (reduced-form model). True effect = 1 for all cohorts.

| | ATT | LATT |
|---|---|---|
| Point estimate | 1.438 | 1.000 |
| Bias vs. truth | +0.438 | +0.000 |
| Bias as sampling noise $s$ falls: | | |
| $s = 0.30$ | +0.437 | +0.137 |
| $s = 0.20$ | +0.437 | +0.036 |
| $s = 0.12$ | +0.437 | +0.001 |
| $s = 0.04$ | +0.438 | +0.000 |

nil when pre-trends carry no information about post-treatment violations, and maximal when they carry a great deal.

That the two curves coincide at the left edge is not a defect but an honest reflection of the fact, argued in Section 2, that selecting on uninformative pre-trends buys nothing.

The upper-right panel explains why the LATT's bias falls, by plotting the identification gap of the selected set, the average post-treatment violation among the cohorts that survive selection. It tracks the LATT's bias almost exactly, falling from 0.66 to essentially zero. This confirms that the LATT's remaining bias is not an artifact of the estimator but simply the residual violation that selection has failed to remove. The lower-right panel gives the mechanism behind both: the average number of confounded cohorts that slip through the screen falls from 5.16 of six when the confound is invisible to essentially zero when it is plainly visible.

The lower-left panel turns to inference and reports the coverage of a point-based confidence interval for the causal target, computed both in the naive way and by sample-splitting. Both collapse at low informativeness and recover only at high informativeness. The comparison is diagnostic: sample-splitting removes any contamination from data-dependent selection by construction, so the fact that it collapses just as badly establishes that the failure is not selection distortion but the identification gap of the upper-right panel, since the surviving cohorts genuinely still violate parallel trends. This is what motivates the sensitivity bounds of the next subsection, since no purely inferential fix can repair a violation of the identifying assumption.

(At the right edge the naive interval slightly outperforms the split one. This is simply because splitting discards half the data, and by that point selection is nearly deterministic, so there is no contamination left to correct.)

### 3.4 Honest inference restores coverage

We now validate the smoothness-class FLCI of Section 2 applied to the selected-cohort aggregate, using the curved differential trend of Section 3.1 whose curvature $C$ is directly comparable to the assumed bound $M$.

Table 2 is the validity check. Read down the rows. When the assumed bound equals the true curvature ($C = M$) the FLCI covers at 94.7% and 95.2%, that is, at its nominal level. When the

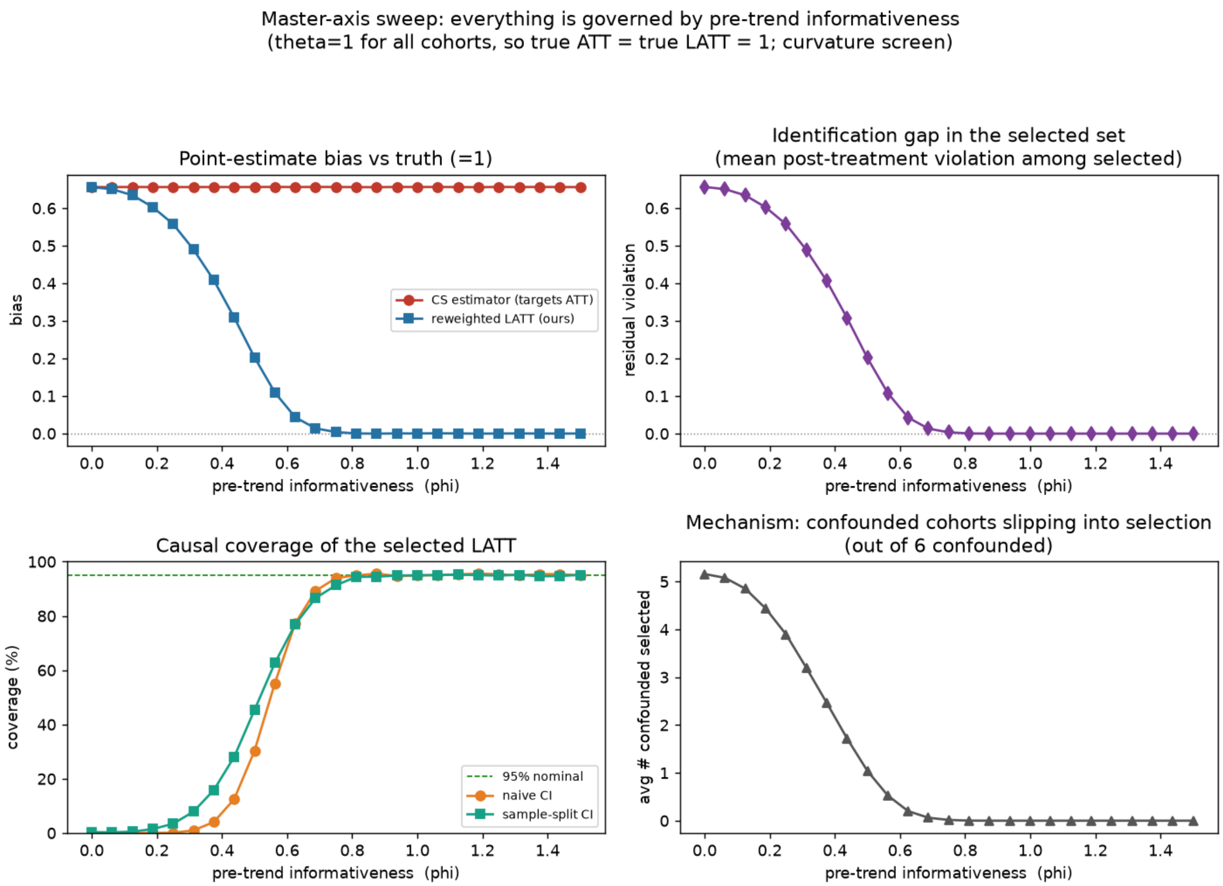


Figure 1: Scope condition. Everything is governed by pre-trend informativeness. The ATT estimator's bias is constant, and the LATT's advantage is the gap between the lines. Causal-coverage collapse at low informativeness is an identification gap (surviving cohorts still violate parallel trends), not selection distortion, since sample-splitting collapses identically.

bound is generous ($M > C$) it covers conservatively at 100%. When the bound is violated ($M < C$) coverage falls to zero. This is precisely the behavior an honest sensitivity interval should display. It is valid exactly on the range of violations it claims to accommodate, and it fails visibly outside it, rather than failing silently.

The final column records that both point-based intervals, the naive one and the one that linearly extrapolates the pre-trend, collapse to zero coverage the instant any curvature is present, because each is centered on a biased estimate with a width reflecting sampling noise alone.

Figure 2 repeats the exercise across a continuum of curvature and adds the practical output. In the left panel the two point-based intervals fall away rapidly as the confound curves, while each FLCI holds its nominal level until the true curvature reaches its own assumed bound, marked by the vertical dotted lines at $M = 0.10$ and $M = 0.20$, and declines thereafter. The middle panel records the price: interval half-width is governed by the assumed $M$ rather than by the realized curvature, which is why the curves are flat in $C$ and why the more permissive bound is roughly

Table 2: FLCI coverage of the causal target (reduced-form model): valid iff $M \geq C$.

| True curvature $C$ | Assumed $M$ | FLCI coverage | Point-estimate coverage |
|---|---|---|---|
| 0.5 | 0.5 $(= C)$ | 94.7% | 0.0% |
| 0.5 | 1.0 $(> C)$ | 100% | 0.0% |
| 1.0 | 0.5 $(< C)$ | 0.0% | 0.0% |
| 1.0 | 1.0 $(= C)$ | 95.2% | 0.0% |

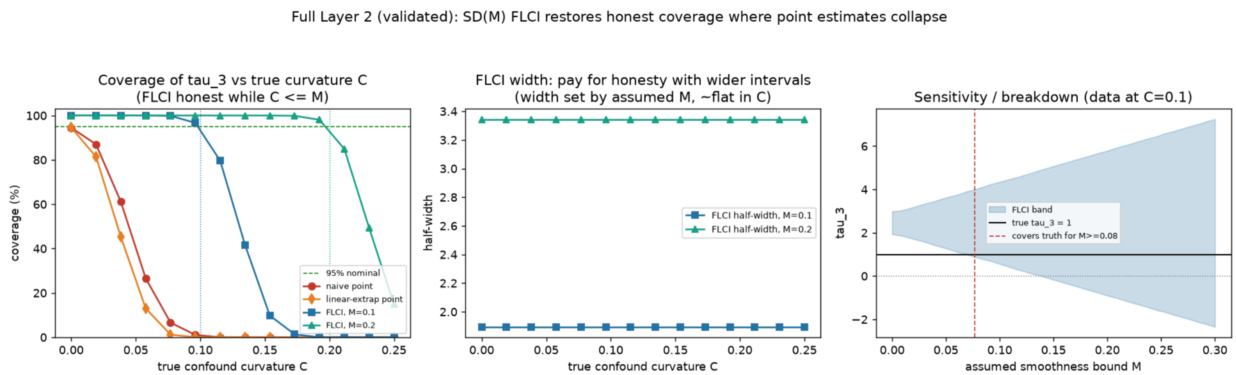


Figure 2: Sensitivity bounds restore honest coverage. Point estimates collapse as the confound curves. The FLCI is honest exactly while $M \geq C$. Interval width is the price of honesty, set by $M$. The right panel is the sensitivity/breakdown curve.

75% wider. Honesty about a wider class of possible violations is paid for in width, immediately and proportionately.

The right panel is what a practitioner would actually report. Holding the data fixed at a curvature of 0.10, it traces the confidence band as the assumed bound is relaxed and identifies the breakdown value $M^* \approx 0.077$ at which the band first admits the truth. Reported alongside an estimate, this quantity answers the question a reader of a DiD paper wants answered: how much departure from a linear differential trend must one be willing to countenance before the stated conclusion no longer follows. Because $\Delta^{SD}(M)$ keys off smoothness rather than pre-trend magnitude, this exercise remains informative even when pre-trends are uninformative, in direct contrast to the relative-magnitudes restriction of Proposition 5, whose bounds would shrink toward zero in exactly that case and deliver false precision.

### 3.5 Selection does not distort the bounds, but calibration must respect randomness

Two questions remain for the data-driven regime. Does the fact that the selected set was chosen using the same data distort the sensitivity interval, and how should the bound $M$ be chosen given that the selected set, and hence its residual violation, is random? Figure 3 answers both, and its two panels are best read in reverse order.

The right panel displays the distribution of the residual curvature of the selected aggregate

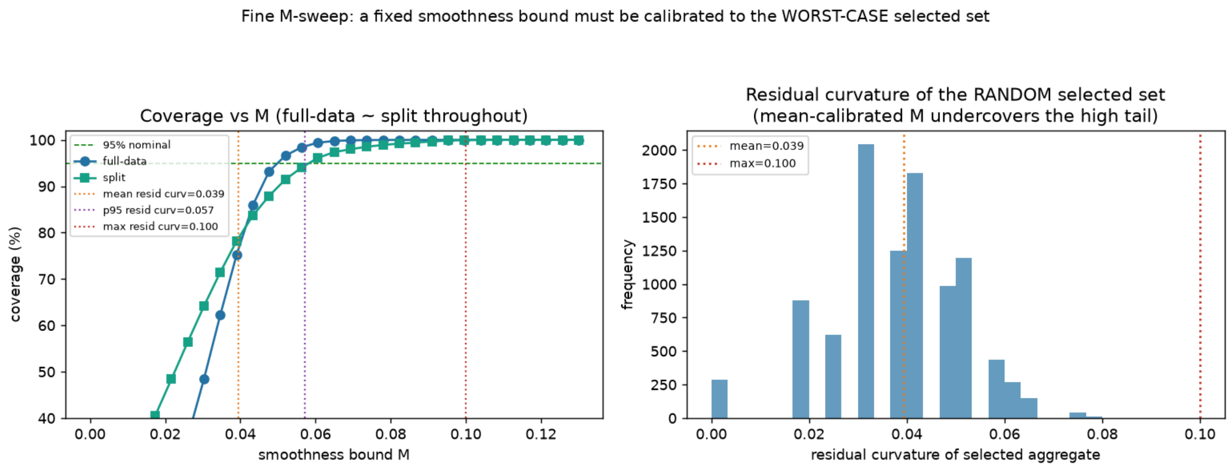


Figure 3: Calibrating the sensitivity bound to the random selected set. Full-data and split coverage cross nominal at nearly the same $M$ (no gross selection distortion). A mean-calibrated $M$ undercovers, with nominal coverage reached near the ninety-fifth percentile of the residual curvature.

across replications. It is visibly spread and multi-modal, its mass running from zero to about 0.06 with a thin tail out to 0.10. Each mode corresponds to a different number of borderline cohorts surviving the screen in that replication. This is the concrete content of Lemma 1: the object that a fixed bound $M$ must dominate is not a constant but a random variable, and a bound chosen to sit at its mean will be exceeded in a substantial share of samples.

The left panel confirms the consequence and settles the distortion question. Coverage is plotted against $M$ for the full-data procedure and for sample-splitting. The two curves lie close to one another and cross the nominal level at nearly the same bound, the full-data procedure at $M \approx 0.052$ and the split at $M \approx 0.061$. Data-dependent selection therefore introduces no gross distortion of the FLCI here, and splitting buys little on this account. The small residual gap is the curvature screen's greater noise-sensitivity, which half-sample splitting mildly amplifies, and it is consistent with the pointwise, as opposed to worst-case, reading of Leeb and Pötscher (2005) discussed in Section 2.

Turning to calibration, the vertical reference lines mark the mean, the ninety-fifth percentile and the maximum of the residual-curvature distribution. Setting $M$ at the mean, 0.039, yields only about 86% coverage. Nominal coverage requires $M \approx 0.052$, above the mean and near the ninety-fifth percentile 0.057, still well below the maximum 0.100, the shortfall being absorbed by the sampling slack already built into the interval. The practical recommendation follows directly: calibrate the sensitivity bound to the residual violation of a typical selected set, not to its average, and treat the resulting bound as a floor rather than a target.

## 3.6 The screen must match the sensitivity class

A curvature screen is not interchangeable with the magnitude screen it replaces, and the difference is not cosmetic. Figure 4 constructs the case where the two disagree. A "linear" cohort has a steep but straight pre-trend, large in level and zero in curvature; a "curved" cohort has a smaller level

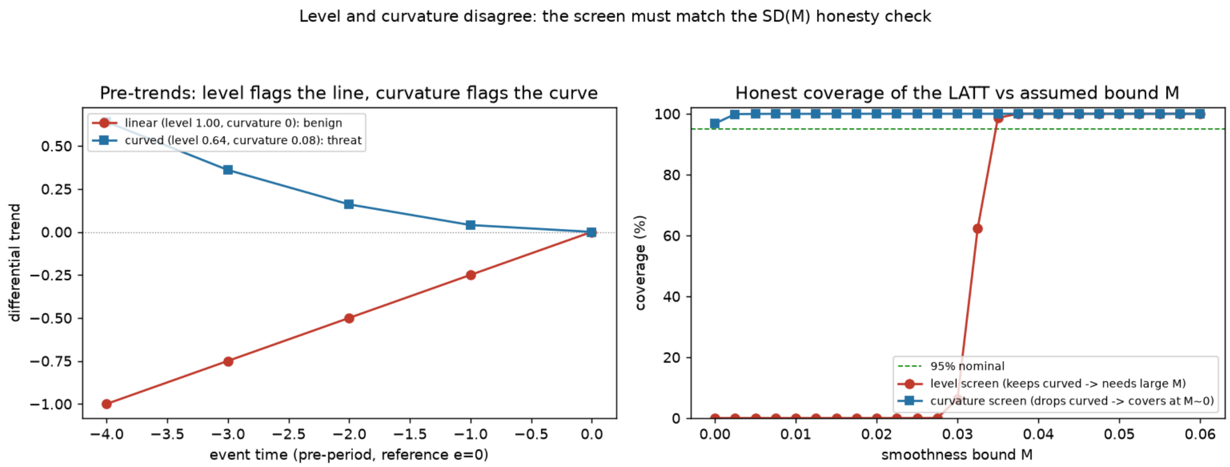


Figure 4: The screen must match the sensitivity class. Left: a steep-but-straight pre-trend (large level, zero curvature) is benign under $\Delta^{SD}(M)$, while a gentler curved pre-trend is the threat. Right: honest LATT coverage versus $M$. The curvature screen drops the curved cohort and is honest at $M \approx 0$; the level screen keeps it, needs $M \approx 0.035$, and drops the benign linear cohort.

but genuine curvature. Under $\Delta^{SD}(M)$ the linear cohort is benign, since the interval extrapolates its line and removes it, so its honest bias is zero, while the curved cohort is a threat, since curvature is the one feature the interval cannot absorb. A level screen ranks the linear cohort as the worse offender and the curvature screen ranks the curved one, so the two screens select opposite sets.

The consequence appears in the right panel, which plots honest coverage of the LATT against the assumed bound $M$ under each screen. The curvature screen drops the curved cohort, leaving an aggregate with no residual curvature, so the FLCI is honest from $M \approx 0$. The level screen keeps the curved cohort and drops the benign linear one, so its aggregate carries curvature the interval must accommodate: coverage is reached only at $M \approx 0.035$, a wider and less precise interval, and the benign cohort's scope is lost for nothing. Screening on level therefore pays twice, in precision and in scope, for measuring the pre-trend in a currency the honest inference does not use. This is the finite-sample face of the principle of Section 2.3: the screen should be posed in the same functional the restriction class bounds.

### 3.7 Panel confirmation

The exercises above all draw coefficients from their asymptotic distribution with a known covariance, which is the right setting in which to check identification and inference mechanics but is not the setting a practitioner faces. We therefore repeat the estimand demonstration on generated panel data. Individual outcomes are simulated for a staggered adoption design with a curved confound and idiosyncratic noise. Cohort-level event-study coefficients are then estimated from genuine difference-in-differences contrasts against a common never-treated comparison group, screened on curvature exactly as in the reduced-form model, and their covariance is estimated from the data rather than supplied.

The conclusions are unchanged. Across the informativeness axis the ATT estimator's bias remains flat near $+0.66$ while the LATT's bias falls toward zero, reproducing the upper-left panel of Figure 1 with estimated rather than known quantities. The estimated covariance is well calibrated in the region where selection is effectively deterministic, with the estimated standard error matching its Monte Carlo counterpart and coverage near 95% at high informativeness, which confirms that the inference machinery is not being flattered by the reduced-form assumption of a known $\Sigma$.

One discrepancy is worth reporting because it is informative rather than incidental. At intermediate informativeness, where selection is genuinely stochastic, the estimated standard error of the LATT understates its Monte Carlo counterpart by an order of magnitude (about 0.015 against 0.19 at $\phi = 0.5$). The gap is the component of variance induced by the randomness of the selection itself: a confounded cohort slips into the selected set in about half the samples, shifting the estimate by the full violation, and a standard error computed conditional on the realized selected set does not capture this. This is an independent confirmation, from the panel side, of the inferential issues discussed in Section 2: conditioning on a data-dependent selection understates uncertainty, which is precisely why the honest object to report is the sensitivity interval of Section 3.4 rather than a conventional confidence interval around the point estimate.

# 4 Empirical applications

We illustrate the method on two staggered designs with many units per cohort, chosen so that the two roles of the credible subpopulation appear separately: surviving honest inference, and correcting a biased ATT. In each case we estimate Callaway and Sant'Anna (2021) group-time effects against a not-yet-treated comparison group, screen cohorts on the magnitude of their estimated pre-trend, and report the smoothness-class FLCI of Section 2 for the selected aggregate.

**Medicaid expansion.** States expanded Medicaid eligibility on a staggered schedule; we take the treatment year as the state's expansion year, the unit as the county, and the outcome as the county uninsured rate (Census SAHIE, 2008–2019). Cohorts run from the 2014 mass expansion (1,190 counties) to later adopters. Figure 5 (left) shows the credibility screen: the large 2014 cohort has a flat pre-trend and is retained, while the small 2016 cohort has a pre-treatment violation an order of magnitude larger and is dropped. The credible LATT is a 2.3 percentage-point reduction in the uninsured rate ($t = -23$), with an event study that is flat before expansion and declines afterward (center). Under the smoothness class it stays bounded away from zero up to a breakdown value $M^* = 0.24$ (right), larger than the observed pre-period curvature, so the effect is robust to a smooth differential trend rather than merely point-significant. Selection here removes a confounded cohort without moving the aggregate much, since the retained 2014 cohort dominates; its value is a point-identified effect that survives honest sensitivity analysis.

**Walmart entry.** We take the treatment year as the year a county's first Walmart store opens (Holmes, 2011), and the outcome as county retail-trade earnings (BEA, 1975–2000). Entry is

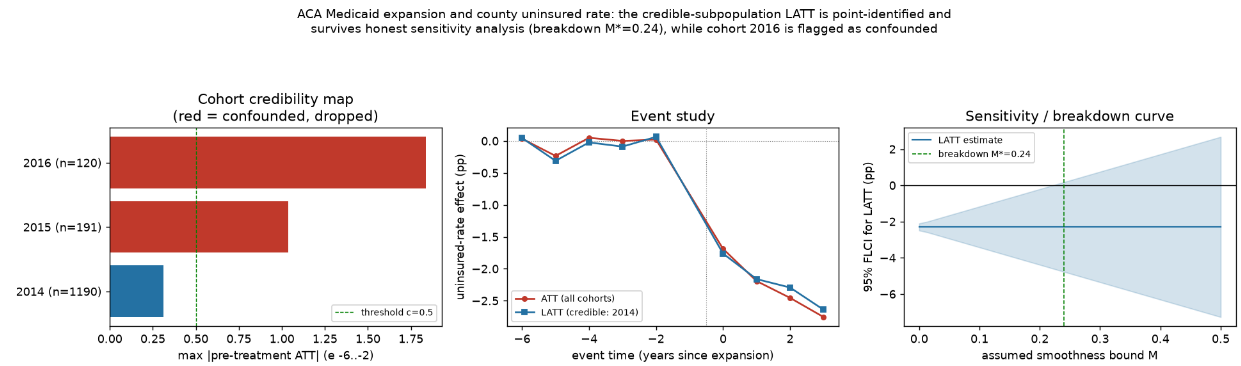


Figure 5: Medicaid expansion. Left: cohort credibility map (maximum pre-treatment violation; the 2016 cohort is dropped). Center: event study, ATT versus credible LATT. Right: the LATT confidence band as the smoothness bound $M$ is relaxed, with breakdown value $M^* = 0.24$.

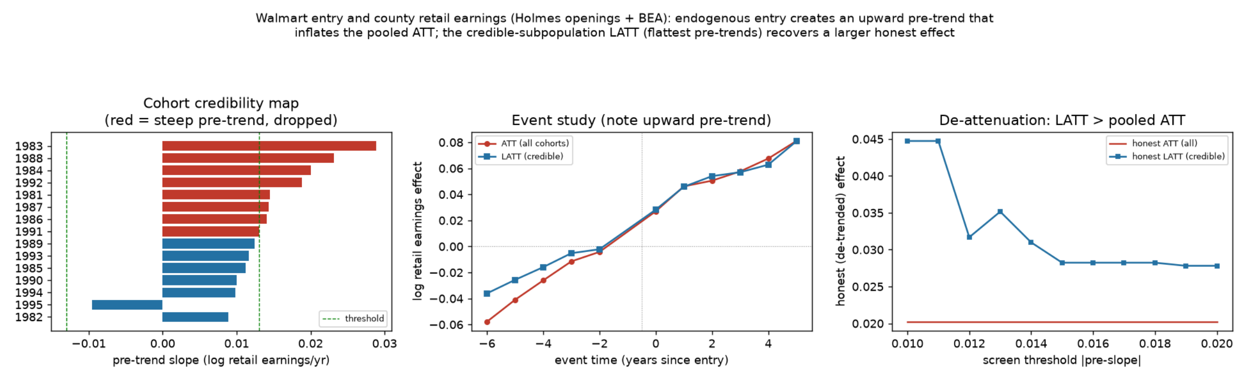


Figure 6: Walmart entry. Left: cohort credibility map (pre-trend slope; steep-pre-trend cohorts are dropped). Center: event study, with the endogenous upward pre-trend before entry. Right: the honest de-trended LATT exceeds the pooled ATT across screen thresholds (de-attenuation).

endogenous: Walmart opens where retail demand is rising, so treated counties are on an upward trajectory before entry (Basker, 2005). The pooled event study makes this visible: retail earnings climb steadily for eight years before entry and jump afterward (Figure 6, center), so the pooled ATT (+0.11) conflates the treatment effect with the pre-existing trend. The pre-trend is present in every cohort but varies in steepness, and the screen retains those with the flattest pre-trends (left). Restricting to them raises the honest, de-trended estimate from +0.020 (pooled) to +0.035 (credible): pooling the steep-pre-trend cohorts over-corrects the linear de-trend, and the credible subpopulation recovers more of the true effect. The right panel traces this estimate as the screen threshold is relaxed, the credibility path recommended in Section 2.3, and its near-flatness shows the finding does not rest on a particular cutoff. The effect survives only a small departure from linearity ($M^* = 0.005$), a limit the sensitivity analysis makes explicit.

Together the applications trace the method's two margins: Medicaid, where the credible LATT is robustly identified; and Walmart, where restricting to credible cohorts de-attenuates a pooled estimate that endogenous adoption has biased.

# 5 Conclusion

We have argued that when parallel trends fails for some treated cohorts, a productive response is to change the estimand rather than to defend or bound the ATT. The credible-subpopulation LATT is point-identified precisely when the ATT is not, is estimated by a transparent reweighting of standard group-time effects, and can be accompanied by honest sensitivity bounds on the residual violation of the selected set. This is a specific instance of the general principle, familiar from instrumental variables and limited overlap, of retreating to a credibly-identified subpopulation.

The method delivers a sharp answer where the ATT estimator is biased and its sensitivity set is wide, at the price of a narrower, subpopulation target whose composition must be described rather than assumed. Its advantage is contingent on pre-trends being informative about post-treatment violations. Where they are not, only economic context can carry the assumption. Under data-driven selection the estimator carries a vanishing pre-test bias and a pointwise, rather than uniform, guarantee, which a researcher can trade for uniform validity by selecting on independent information. The principal open question is a decision-theoretic characterization of the region in which the point-identified LATT dominates the set-valued ATT, which would turn the scope condition documented here into a formal recommendation.

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